# Wind Farm Layout Optimization

*A Comparative Study of Heuristic and Machine Learning Approaches*

| | | | |
|---|---|---|---|
| **50** | **4 km²** | **7 yrs** | **4** |
| Turbines | Farm Area | Wind Data | Algorithms |

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# The Wake Effect Problem

### The Cinema Analogy

A tall person in the front row blocks the view for everyone behind. In a wind farm, an upstream turbine "steals" the wind — leaving a slower, turbulent wake that reduces power for downstream turbines.

### Why It Matters

A suboptimal layout can cost 5–15 % of Annual Energy Production (AEP) — eliminating the business case for the entire project.

### Perimeter Constraint

All 50 turbines within 50 m of farm edge

### Proximity Constraint

Min. 400 m between any two turbines

### Objective

Maximise AEP across 540 wind instances

# Wind Resource Analysis

| | | | |
|---|---|---|---|
| **7** | **540** | **>0.98** | **178/540** |
| Years of data (2007–2017) | Wind instances (36 dir × 15 speed) | Cross-year Pearson correlation | Bins covering 80 % of probability |

- Dominant wind directions: 200°–260° (South-Southwest) carry the largest AEP contribution.
- Top 5 directions (200°, 210°, 260°, 220°, 250°) account for the majority of energy yield.
- Wind speeds in 6–12 m/s range are most frequent and overlap peak thrust coefficient region.
- Year-to-year wind distributions are highly stable (Pearson $r > 0.98$), enabling a single ensemble-averaged probability map.

# Turbine Specs & Wake Model

## Turbine Specifications

| | |
|---|---|
| Rated Power | **3.0 MW** |
| Rotor Diameter | **100 m** |
| Hub Height | **100 m** |
| Cut-in Speed | **3.5 m/s** |
| Rated Speed | **15 m/s** |
| Cut-out Speed | **25 m/s** |
| Wake Decay Const. | **kw = 0.05 (offshore)** |

## Jensen PARK Wake Model

The speed deficit at downstream distance x:

$$\Delta V/V_\infty = (1 - \sqrt{(1-C_T)}) \times (D / (D + 2 \cdot kw \cdot x))^2$$

- **ΔV:** Wind speed reduction at target turbine
- **$V_\infty$:** Free-stream wind speed
- **$C_T$:** Thrust coefficient (from power curve)
- **D:** Rotor diameter (100 m)
- **kw:** Wake decay constant (0.05 offshore)

⚡ At minimum spacing (400 m) with $C_T \approx 0.79$, the wake deficit reaches ≈ 27.6% — making turbine alignment along dominant wind directions extremely costly.

# Algorithm Selection & Pipeline

| Algorithm | Score | Handles Wake | Scalability | Speed |
|---|---|---|---|---|
| Greedy Heuristic | High | Good | Excellent | Very Fast |
| Genetic Algorithm | ★ 64/80 | Good | Good | Moderate |
| XGBoost Surrogate | High | Proxy | Good | Fast (after train) |
| Reinforcement Learning | High | Proxy | Moderate | Slow (training) |
| MINLP (exact) | Low | Exact | Poor (>20) | Very Slow |

## Recommended 6-Step Pipeline

1. Ensemble-avg 540-bin probabilities
2. Seed feasible grid population
3. Run algorithms
4. Repair constraint violations
5. Local refinement (hill-climbing)
6. Verify AEP & constraints

# Method 1 — Greedy Algorithm

*Intuition: Build the layout step-by-step — at each step, place the next turbine at the position that gives the greatest immediate AEP gain. Then polish with local refinement.*

## Algorithm Steps

1. Discretise farm into 400 m grid → candidate locations
2. Start with empty layout
3. For each turbine slot: evaluate AEP for every valid candidate
4. Place the turbine with highest marginal AEP gain
5. Repeat until 50 turbines are placed
6. Refine: move each turbine ±200 m (step 20 m), keep if AEP improves
7. Repeat refinement up to 10 passes

## Results

| | |
|---|---|
| Greedy construction | 506.02 GWh |
| After local refinement | 526.44 GWh |
| Improvement | +20.42 GWh (+4.0%) |

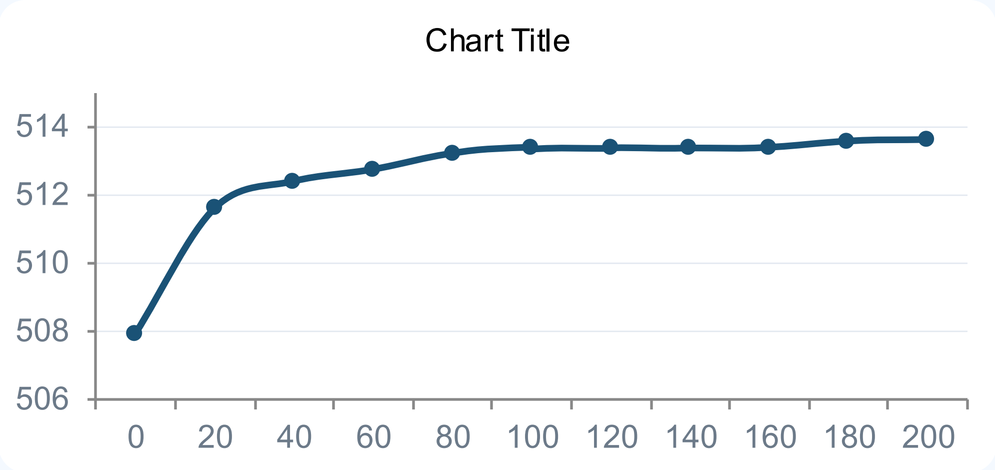
**526.44**

**GWh / year • Best method overall**

Deterministic, runs in ~1 minute, and reaches the theoretical optimum — the most powerful baseline.

# Method 2 — Genetic Algorithm

*Mimics biological evolution — layouts breed, mutate, and compete over 200 generations. No domain knowledge required.*




| Stage-by-Stage Results | |
|---|---|
| Random start (Gen 0) | **507.91** |
| GA best (Gen 200) | **513.64** |
| **GA + Local Refinement** | **523.67** |
| Baseline (Greedy) | **526.44** |

- GA alone adds +5.73 GWh over a random start purely through natural selection.
- Local refinement contributes a further +10.03 GWh — a powerful hybrid approach.
- Final result is only 2.76 GWh (0.52 %) below the domain-expert Greedy baseline.

# Method 3 — XGBoost Surrogate

*Train a fast ML model to approximate AEP, then use it to screen 200,000 layouts in seconds before evaluating the best with the true simulator.*

**1**
**Grid**
10×10 candidate grid (400 m spacing)

**2**
**Train**
5,000 random layouts labelled w/ AEP

**3**
**Screen**
Score 200,000 layouts via XGBoost

**4**
**Top-100**
True AEP eval for top candidates

**5**
**Refine**
Local hill-climb ±100 m, 10 passes

**1.55 GWh**
XGBoost RMSE (validation)

**500.60**
Best surrogate layout AEP (GWh)

**520.54**
After local refinement (GWh)

**−1.12 %**
Gap vs Greedy baseline

After one-time training on 5,000 samples, the surrogate screens 200,000 layouts in seconds — a practical approach when the true evaluator is expensive. Final AEP is only 1.12 % below Greedy.

# Method 4 — Reinforcement Learning (DQN)

*A DQN agent builds the layout sequentially — placing one turbine per step, guided by a neural-network surrogate trained on hand-crafted layout features.*

### STATE

Coordinates of turbines already placed in current episode

### ACTION

Choose next turbine location from 400-point discrete grid

### REWARD

Surrogate-predicted AEP of the completed layout

**Training Configuration**

| | |
|---|---|
| Episodes | **500** |
| Surrogate training samples | **500** |
| Exploration (ε) | **1.0 → 0.05** |
| Grid | **20×20 (200 m spacing)** |

**Performance Results**

| | |
|---|---|
| RL (surrogate reward) | **488.83 GWh** |
| After local refinement | **519.61 GWh** |
| Improvement | **+30.78 GWh (+6.3%)** |
| Gap vs Greedy | **−6.83 GWh (−1.30%)** |

Trained without greedy heuristics and with only 500 episodes — yet reaches 519.61 GWh. Scaling training will close the remaining gap.

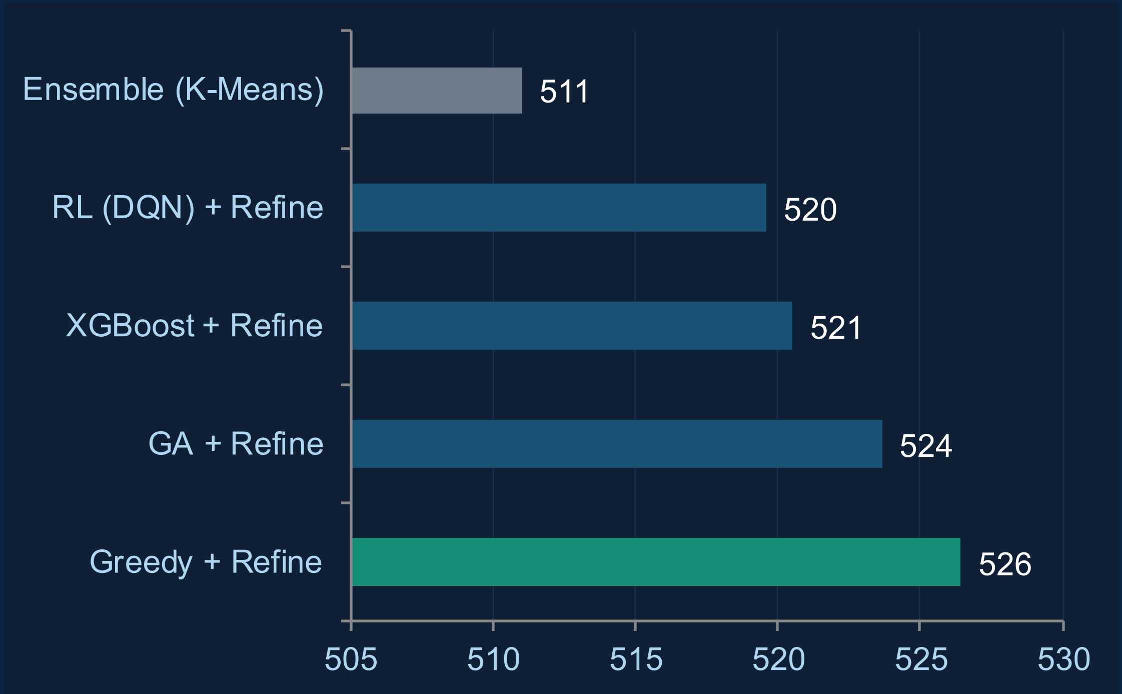
# Method 5 — Ensemble & Clustering

*Combine the wisdom of all four methods into a single robust layout using Hungarian matching and group-centroid computation.*

1. Load all 4 layouts
2. Hungarian algorithm match
3. Form 50 groups of 4
4. Compute centroids
5. Apply constraints + gap-fill

| Metric | Value | Notes |
|---|---|---|
| Groups formed | **50** | One per turbine slot |
| Intra-group spread (min) | **54.5 m** | Tightest agreement |
| Intra-group spread (max) | **1,306.7 m** | Largest divergence |
| Intra-group spread (mean) | **387.5 m** | Below 400 m spacing |
| Centroids after greedy | **45** | 5 groups too close |
| Gap-filled turbines | **5** | Farthest-first fill |
| Boundary constraint | **PASS ✓** | All within [50, 3950] m |
| Proximity constraint | **PASS ✓** | Min pair dist 407.5 m |

# Performance Comparison — All Methods




| # | Method | AEP | Gap |
|---|---|---|---|
| ★ 1 | **Greedy + Refine** | **526.44** | — |
| 2 | GA + Refine | 523.67 | −0.53 % |
| 3 | XGBoost + Refine | 520.54 | −1.12 % |
| 4 | RL (DQN) + Refine | 519.61 | −1.30 % |
| 5 | Ensemble (K-Means) | 511.05 | −2.92 % |

**Greedy + Local Refinement is the recommended final layout — 526.44 GWh, all constraints satisfied, deterministic and reproducible.**

# Key Insights

## 01 Greedy Works Best

Step-by-step placement with local refinement reaches 526.44 GWh — the highest of all methods. Deterministic, fast (~1 min), and physically motivated.

## 02 GA is a Strong Runner-Up

Without any wind-physics knowledge, the Genetic Algorithm reaches 523.67 GWh (only 0.53 % below Greedy) — highly competitive for a general-purpose solver.

## 03 ML Methods: Promising, Data-Hungry

XGBoost (520.54 GWh) and RL (519.61 GWh) are within 1.3 % of Greedy even with small training sets. More data and longer training will close the gap.

## 04 Ensemble Averaging Hurts

Merging four layouts into one consensus (511.05 GWh) blurs fine-tuned positions. When one method clearly wins, ensembling coordinates is counterproductive.

## 05 Local Refinement Always Pays Off

Every method gained 10–30 GWh (~2–6 %) from the same post-processing: move each turbine ±200 m and keep if AEP improves. An essential step in any pipeline.

# Future Directions & Conclusion

## Future Directions

- Scale ML surrogate training data to reduce the gap with the Greedy baseline
- Explore NSGA-II and other multi-objective GA variants for robustness
- Investigate more complex wake models (e.g., Gaussian) where ML surrogates become even more attractive
- Apply a weighted ensemble — give higher influence to the best-performing method rather than simple averaging
- Trial Random Forest and other surrogate architectures
- Extend to offshore terrain and variable hub-height configurations

## Recommendation

**Greedy**
**+ Refinement**
**526.44 GWh**

All constraints ✓
Deterministic ✓
Fast (~1 min) ✓
Reproducible ✓

This project demonstrates that a simple, physics-motivated heuristic — combined with local refinement — outperforms complex ML approaches for wind farm layout optimisation under the Jensen PARK model.